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User Guide & Component Testing Manual: Multi-Store Facilitator & Outsourced Delivery PoC

1. Quick Access URLs (Live on Vercel Cloud & Local Services)

Service Name	Live Vercel Production URL	Local URL	Purpose
1. Multi-Store Marketplace Facilitator PoC	https://egypt-ecommerce-poc.vercel.app	http://localhost:4000	Main application with **6 dedicated workspaces**: Client Storefront, My Orders & Tracking, Mobile App, Partner Store Portal, Outsourced Delivery Hub, and Facilitator Control Tower.
2. Egyptian Fintech & Outsourced 3PL Mockup Server	https://egypt-ecommerce-poc.vercel.app/mock-dashboard	http://localhost:4001	Sandbox simulating **Paymob Split**, **FawryPay**, **valU**, **WhatsApp COD OTP**, **Bosta Store-to-Client Delivery**, and **Cash Settlement**.
3. Architecture & 1st-Year Liability Study	https://egypt-ecommerce-poc.vercel.app/study	http://localhost:8080	Comparative study of the Top 5 platforms and 1st-Year Liability Calculator.

2. Step-by-Step Guide to the 6 Dedicated Workspaces on `http://localhost:4000`

Open **http://localhost:4000** in your browser and press `Ctrl + R` (refresh) to load the new **Eye-Friendly Light/Dark UI**.

- 🌞 **Light / 🌙 Dark Theme Switcher:** The platform defaults to a crisp, high-contrast **Light Mode** that is easy on the eyes, with a 🌙 **Dark Mode / 🌞 Light Mode** button in the top-right header.
- 🌐 **Bilingual English & Arabic RTL:** Switch between English and native Arabic layout with one click on `🌐 العربية`.

Tab 1: `🛒 1. Client Storefront` (Spacious Catalog + 3-Step Checkout)

1. **Filter by Product Category:** Click any category pill:

- `🌐 All Categories`, `📱 Electronics & Mobile`, `👘 Egyptian Fashion & Cotton`, `☕ Home & Coffee Appliances`, `✨ Beauty & Skincare`, `👟 Sports & Footwear`.

2. **Filter by Partner Store:** Use the **Filter by Partner Store** dropdown to view products from a specific store (*Cairo Tech Hub*, *Giza Cotton Atelier*, *Alex Glow & Home*).

3. **Multi-Store Cart Split:** Add items from multiple partner stores; notice automatic Bosta multi-pickup dispatch, facilitator commission calculation, and net store payout preview.

4. **Place Order:** Click 🔒 **Place Order & Dispatch to Partner Store(s)** and complete the payment simulation (Paymob 3DS, Fawry Kiosk Cash, valU OTP, InstaPay, or WhatsApp COD OTP).

5. **Direct Tracking Link:** Click 📦 **Track My Orders** in the hero banner or order modal to instantly jump to your orders dashboard.

Tab 2: `📦 2. My Orders & Tracking` (Active Shipments & Historical Closed Archive)

1. **Click `📦 2. My Orders & Tracking` in the top navigation bar.**

2. **High-Level KPI Summary:**

- **Total Client Orders:** Lifetime order count.
- **Active & In-Transit (Pulsing Cyan):** Orders currently undergoing store packaging or transit with Bosta.
- **Completed & Closed (Emerald):** Delivered, settled, or archived orders.
- **Total Purchase Value (EGP):** Cumulative fulfilled purchases.

3. **Active vs. Closed Scope Filtering:**

- Click 🌟 **All Orders**, 🚚 **Active & In-Transit**, or ✅ **Completed & Closed** to filter orders instantly.

4. **Live Search Bar:**

- Type in the search box to filter live by **Order Number** (e.g. `EG-36711-31`), **Recipient Phone** (`010...`), or **Partner Store Name**.

5. **5-Stage Visual Progress Stepper:**

- Each order card displays a visual milestone pipeline (`1. Order Placed` → `2. Payment Verified & Split` → `3. Store Packaged` → `4. Out for Delivery` → `5. Delivered & Settled`) with status indicators, percentages, and timestamps.

6. **Partner Store Sub-Manifests & Bosta AWBs:**

- Inspect each fulfilling partner store branch, merchant packaging state, and click the **Bosta Waybill (`BST-xxxxxxx ↗`)** to open the 3PL sandbox tracker.

7. **Auditable Tax Invoice Modal (`#invoiceModal`):**

- Click 📄 **View / Print Tax Invoice** to display the official commercial invoice featuring NileConnect Tax ID (`492-817-309`), Commercial Register (`184920`), itemized store origins, VAT 14%, and a 1-click 🖨️ **Print / Save PDF** button.

8. **1-Click Support Chat & Cart Reorder:**

- Click 💬 **Inquire with Support** to open the customer support drawer pre-populated with your order number.
- Click 🔁 **Reorder Items** to reload all products into your shopping cart.

Tab 3: `📱 3. Mobile App` (iOS & Android Shopping & Live Push Stream)

1. Click 📱 **3. Mobile App** in the top bar.

2. Switch between 🍎 **iOS (Apple Pay)** and 🤖 **Android (Google Play)**.

3. Tap product categories, add items to cart, and tap **Pay / Place Mobile Order**.

4. Watch the **live push notification banner** update in real-time as stores pack your items and couriers pick them up.

Tab 4: `🏪 4. Partner Store Portal` (Dedicated Screen for Partner Stores)

1. Click 🏪 **4. Partner Store Portal** in the top bar.

2. Switch between onboarded stores (*Cairo Tech Hub*, *Giza Cotton Atelier*, *Alex Glow & Home*).

3. Adjust inventory (`+5 Stock` / `-1`) or publish new products under any category.

4. In the **Incoming Sub-Orders to Pack** table, click 📦 **Mark Packaged for Bosta** to ready parcels for 3PL pickup.

Tab 5: `🚚 5. Outsourced Delivery Hub` (Dedicated Screen for Bosta 3PL Delivery)

1. Click 🚚 **5. Outsourced Delivery Hub** in the top bar.

2. View direct **Store Pickup Origin → Client Drop-off Destination** manifests.

3. Follow the **Guided Next-Step Pipeline Button**:

- `Step 1: Store Packed` → `Step 2: Bosta Picked Up` → `Step 3: Out for Delivery` → `Step 4: Delivered` → `💸 Step 5: Settle Cash & Commission`.

4. Each transition triggers HMAC-signed webhooks, updates Bosta waybills, and notifies clients.

Tab 6: `⚙️ 6. Facilitator Control Tower` (360° Visibility, Commercial Policy Engine & Support Desk)

1. Click ⚙️ **6. Facilitator Control Tower**.

2. **8 Executive KPI Cards:** Realized vs. pending commissions, store escrow payouts, Egypt VAT ledger (14%), and webhook health.

3. **Commercial Policy Engine:**

- Set Commission %, Egypt VAT %, VAT Mode (`INCLUSIVE` vs. `ADDITIVE`), Delivery Fee Override, COD Fee, and Settlement Cycle across **ALL Stores** or a **Specific Store**.

4. **Governorate Delivery Rates & Bulk Cash Settlement:**

- Adjust Bosta rates per governorate and settle all eligible delivered sub-orders in 1 click.

5. **Customer Support Live Desk:**

- Review incoming Client support tickets (`SUP-xxxx`), reply live to shoppers, or mark tickets resolved.

6. **Master Tables & Webhooks Inspector:**

- Inspect all orders, deliveries, inventory items, and HMAC-SHA512 webhook logs.

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3. Running the Automated 11-Stage End-to-End (E2E) Test Suite

Run the full E2E lifecycle test suite anytime in PowerShell:

```
python "C:\Users\skamal\.gemini\antigravity\scratch\egypt-ecommerce-poc\test_poc.py"
```

Verified E2E Lifecycle Stages (`11/11 Passing`)

- **E2E Stage 1 (Store Adds Product):** Partner Store (`Cairo Tech Hub`) publishes a new product (`Samsung Galaxy Watch 7 Ultra` , `24,000 EGP` , `10` stock) into Category `Electronics & Mobile` .
- **E2E Stage 2 (Client Buy & Payment):** Client buys `2x` Watch (`Store 1`) + `1x` Giza Hoodie (`Store 2`), verifies commission split (`4,014 EGP` Facilitator / `45,436 EGP` Stores), approves Paymob 3DS, and verifies permanent store stock deduction (`10 -> 8`).
- **E2E Stage 3 (Store Fulfillment):** Both Partner Stores pack and mark their sub-orders `READY_FOR_COURIER_PICKUP` .
- **E2E Stage 4 (Outsourced 3PL Delivery):** Bosta picks up from both store branches (`PICKED_UP_FROM_STORE` → `OUT_FOR_DELIVERY` → `DELIVERED`) and marks both sub-orders `ELIGIBLE_FOR_SETTLEMENT` .
- **E2E Stage 5 (Cash Settlement & Commission Realization):** Executes `POST /wp-json/wc/v3/deliveries/{id}/settle` , generating HMAC-verified `STL-xxxxxx` settlement records, paying `+44,160 EGP` to Store 1 and `+1,276 EGP` to Store 2, realizing `+4,014 EGP` Facilitator Commission, and marking the order `completed-settled` .
- **E2E Stage 6 (Full COD E2E Cycle):** Verifies WhatsApp OTP (`4455`) → Store Packaging $

ightarrow$ Bosta COD Collection (`24,070 EGP`) → Bosta COD Remittance & InstaPay Store Payout (`22,080 EGP` to Store 1 + `1,920 EGP` Facilitator Fee, final stock `7`).

- **E2E Stage 7 (Fawry 48h Expiry Protection):** Verifies 3 reserved units are automatically returned to the Partner Store's available stock (`7`) when a Fawry reference code expires.
- **E2E Stage 8 (Webhook Security):** Verifies forged HMAC-SHA512 settlement webhooks are rejected with `HTTP 401` .

- **E2E Stage 9 (Commercial Policy Engine — Per-Store & All-Stores):** Verifies applying custom commission (`15%`), `ADDITIVE` Egypt VAT (`+14%`), Free Delivery override (`0 EGP`), and custom COD fee (`25 EGP`) to Store `#1` , then restoring global rules across `all` stores and executing bulk settlement (`POST /wp-json/wc/v3/settlements/settle-bulk`).
- **E2E Stage 10 (Two-Way Live Customer Support Chat):** Verifies Client opening a support ticket on the Storefront (`POST /wp-json/wc/v3/support/tickets`), Facilitator Admin replying from the Support Desk, Client responding, and Admin marking the ticket `RESOLVED` .
- **E2E Stage 11 (Client Order Monitoring & History Archive):** Verifies `/wp-json/wc/v3/orders/track` endpoint aggregation, active vs. closed order partitioning, 5-stage milestone pipelines, Bosta waybills, single order tracking, and live query filtering.

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4. Deploying the Full Environment to Vercel

The project at `C:\Users\skamal\.gemini\antigravity\scratch\egypt-ecommerce-poc` includes `vercel.json` , `api/index.py` , `requirements.txt` , and a Git repository so all 3 services (`:4000` Core Platform, `:4001` Mock Integrations Sandbox, and `:8080` Study Report) deploy under a single production URL: **`https://egypt-ecommerce-poc.vercel.app`** .